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ORIGINAL ARTICLE

What Differentiates Children with ADHD Symptoms Who Do and Do Not Receive a Formal Diagnosis? Results from a Prospective Longitudinal Cohort Study

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Abstract

ADHD diagnoses are increasing worldwide, in patterns involving both overdiagnosis of some groups and underdiagnosis of others. The current study uses data from a national longitudinal study of Irish children ($N = 8568$) to examine the sociodemographic, clinical and psychological variables that differentiate children with high hyperactivity/inattention symptoms, who had and had not received a diagnosis of ADHD. Analysis identified no significant differences in the demographic characteristics or socio-emotional wellbeing of 9-year-olds with hyperactivity/inattention who had and who had not received a diagnosis of ADHD. However, by age 13, those who had held a diagnosis at 9 years showed more emotional and peer relationship problems, worse prosocial behaviour, and poorer self-concept. Further research is required to clarify the developmental pathways responsible for these effects.

Keywords ADHD · Diagnosis · Longitudinal · Ireland · Child wellbeing

Attention Deficit Hyperactivity Disorder (ADHD) is a disorder characterised by high levels of hyperactivity, impulsivity and inattention, which are associated with impairment in academic, social and/or personal functioning. Diagnosis is usually made by a clinician with expertise in child development following clinical interviews with the child and their caregivers, which may be supplemented by standardised rating scales and/or school observations. ADHD diagnoses are increasing worldwide, in patterns that involve both overdiagnosis of some groups (notably, middle-class white males) and underdiagnosis of others (particularly females and members of ethnic and linguistic minorities) [1–3]. Understanding the implications of these distinctive diagnostic patterns requires information about the predictors and outcomes of receiving an ADHD diagnosis in childhood. The current study uses data from a national longitudinal study of Irish children to examine the sociodemographic, clinical and psychological variables that differentiate children with high hyperactivity/inattention symptoms, who do and do not receive a diagnosis of ADHD. It investigates whether receiving a diagnosis is associated with better or worse longitudinal socio-emotional outcomes for children with ADHD symptoms.

Who Receives a Diagnosis of ADHD?

ADHD is one of the most common disorders of childhood, with an estimated prevalence rate of approximately 5% worldwide [4]. Most childhood psychiatric disorders are underdiagnosed, and similar claims are sometimes made for ADHD, particularly in females [5–7]. However, others in the ADHD field have articulated a concern about overdiagnosis [8, 9]. In the US, rates of diagnosis have exceeded the estimated prevalence rate of 5% for at least 20 years, with diagnoses increasing annually [3, 10, 11]. Figures from the 2016 National Survey of Children's Health suggest 9.4% of US 5–17 year olds had received an ADHD diagnosis and 5.2% were actively taking ADHD medication [12]. Although

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North America remains the region with the highest rates of diagnosis, similar patterns are in evidence globally [13, 14].

The discrepancy between epidemiological estimates and actual diagnostic prevalence suggests diagnostic practice in ADHD is suboptimal. Empirical evidence substantiates this proposal. One study showed that a significant minority (17%) of practicing clinicians diagnosed ADHD in a hypothetical case that did not meet diagnostic criteria, with misdiagnosis twice as high when the case was male as female [15]. Symptomatic girls are less likely to receive a diagnosis than boys [16]. There is extensive international evidence showing that the youngest children in a classroom are disproportionately likely to acquire an ADHD diagnosis and/or medication, irrespective of symptom severity [17–25]. Rates of diagnosis are also influenced by geographic and racial factors, with children from ethnic and linguistic minorities less likely to receive a diagnosis [1, 3]. Finally, there are many individual differences (e.g. cultural outlook, subjective judgements about what constitutes 'normal' child activity levels) that may influence the likelihood of teachers suggesting, parents seeking, or clinicians recommending a diagnosis of ADHD.

To date, minimal epidemiological research has investigated the prevalence or sociodemographic distribution of ADHD diagnoses in Ireland. ADHD is the most frequent primary presentation to Irish Child and Adolescent Mental Health Services (CAMHS), present in 31.6% of service-users [26]. However, use of ADHD medications is lower than other advanced economies [14] and some health professionals show negative attitudes to ADHD diagnosis [27]. The current study supplements the international literature on diagnostic patterns by exploring the prevalence and distribution of ADHD diagnoses in an Irish community sample.

What are the Socio-emotional Implications of a Diagnosis of ADHD?

An informed debate about the phenomena of under- and/or over-diagnosis requires an understanding of the implications of receiving an ADHD diagnosis. That is, does a diagnosis change a child's life trajectory in positive or negative ways? This question does not pertain solely to management of core ADHD symptoms. Research shows that receiving a psychiatric diagnosis in early life affects a child's emotional and social wellbeing in numerous diverse ways [28]. An ADHD diagnosis can bring a sense of self-understanding and relief, mitigating blame of the child for undesirable behaviour [29–34]. Yet children can also experience diagnosis as threatening and devaluing their self-image [29, 30, 32, 34, 35]. Similar inconsistencies are found for diagnosis' effects on social relations: an ADHD diagnosis can expose a young person to stigma and bullying [30, 32–34], yet can also foster more understanding from family, teachers and peers [31, 33, 36].

Given these complex and contradictory effects, which can occur within as well as between individuals [28], it is important to understand the net benefits that diagnosis brings to the socio-emotional wellbeing of symptomatic young people. However, most research exploring social and psychological responses to diagnosis is qualitative in design, and therefore unable to weigh up the various positive and negative effects to summarise their cumulative implications for child wellbeing and clinical practice. Additionally, qualitative studies are unable to isolate the extent to which positive or negative effects follow from the diagnostic label itself, rather than the symptoms that presumably precede it [28]. Quantitative research that compares socio-emotional outcomes for young people of equivalent symptomatology, who do and do not receive a diagnosis, is extremely sparse.

Some insight into the repercussions of diagnosis can be derived from the literature on the effects of ADHD treatment, which diagnosis presumably facilitates. There are safe, easily-administered pharmaceutical treatments for ADHD symptoms, which show documented effectiveness [37–39]. Behavioural interventions may also be helpful [40, 41], particularly when used in conjunction with pharmacological treatment [37, 42]. The symptom reduction that treatment facilitates has positive corollaries for global life outcomes. A systematic review of 351 studies covering a range of academic, health, social and behavioural outcomes confirms that untreated ADHD generally leads to poorer outcomes than either treated ADHD or non-ADHD controls [38]. A further review of longitudinal research specifically focused on self-esteem and/or social function outcomes confirms that persons with untreated ADHD are disadvantaged relative to their treated or non-symptomatic peers [43]. While higher intensity of treatment sometimes correlates with more negative outcomes, this is likely because more severely impaired youths, or those with comorbid psychiatric problems, require more intervention [44]. Thus, to the extent that diagnosis facilitates access to effective treatment, the diagnosis should have a positive effect on a symptomatic young person's wellbeing [6].

However, the literature comparing those with treated and untreated ADHD is not an exact proxy for the current paper's question of the outcomes of ADHD *diagnosis*. While treatment and diagnosis will often be co-extensive in real-world contexts, this is not always the case. US data indicate approximately one-quarter of children with active ADHD diagnoses are not receiving any treatment [12], and this figure is likely higher in other countries [45]. Undertreatment may be due to numerous factors, including resource constraints and clinician or parent/child decisions to forego treatment. Further separating the variables of diagnosis and treatment, it is possible that treatment may be offered to symptomatic children without giving a formal diagnosis[1] [46]. While a diagnosis is a prerequisite for accessing treatment in some jurisdictions, in others it is possible for clinicians to trial treatments for hyperactivity symptoms prior to stipulating a formal diagnosis.

The empirical literature contains one cross-sectional study comparing diagnosed individuals with peers who screened positive for ADHD but had not received a diagnosis [47]. Undiagnosed individuals were disadvantaged relative to non-ADHD controls on quality of life, functional impairment and interpersonal difficulties; however, there was no significant difference when compared to the diagnosed ADHD participants [47]. This study departs from the treatment literature in suggesting that clinical identification of ADHD does not necessarily improve socio-emotional outcomes. However, Able et al.'s [47] study was cross-sectional, making it impossible to draw inferences about directionality (i.e. does diagnosis actually produce minimal benefits, or are children who receive a diagnosis more impaired at baseline?). Moreover, the analytic strategy involved multiple uncorrected tests that did not control for important variables such as symptom severity. To date, no longitudinal research compares socio-emotional development for children with diagnosed and undiagnosed hyperactivity/inattention, controlling for relevant clinical and socio-demographic variables.

The Current Study

The current research uses data from *Growing Up in Ireland – the National Longitudinal Study of Children* (GUI) to investigate the distribution and longitudinal socio-emotional implications of ADHD diagnosis in an Irish community cohort sample. The analysis had three objectives:

(1) to establish the socio-demographic profiles of 9-year olds with diagnosed and undiagnosed ADHD symptoms in an Irish community sample;
(2) to extract relevant information about these two cohorts' health, service use and cognitive ability;
(3) to determine if the diagnosed and undiagnosed cohorts differed on longitudinal socio-emotional outcomes, controlling for symptom severity and other relevant variables.

[1] Such a scenario may also occur in research contexts in double-blinded clinical trials, which further confounds the extent to which research on *treatment* outcomes can be generalised to draw conclusions about the outcomes of ADHD *diagnosis*.

Methods

Data Collection

GUI is an ongoing longitudinal study that tracks the physical, psychological and social development of a nationally representative sample of Irish children. The current analysis focuses on a cohort recruited into the study at 9 years old (n = 8568). Two waves of data from this cohort are available, collected when the children were aged 9 (2007–2008) and 13 (2011–2012). GUI identified potential participants using a stratified sampling strategy based on the primary school system and achieved a household response rate of 57%, representing approximately 15% of the total population of 9-year-olds in Ireland [48]. Wave 2 retained 87.8% (n = 7525) of households. Trained interviewers administered questionnaires to children, their primary caregiver (in 97.8% of cases the child's mother) and teachers using both self-complete and computer assisted personal interviewing techniques. Detailed information on GUI's methodological and ethical procedures is available at http://www.growingup.ie/.

Defining Study Groups

This analysis sought to compare 9-year olds who held a formal ADHD diagnosis with those who showed high hyperactivity/inattention symptoms but had not received a diagnosis. To achieve this, two groups were constructed, labelled 'Diagnosed ADHD' and 'Undiagnosed ADHD'.

The Diagnosed ADHD group was composed of children whose parent reported they held a clinical diagnosis of ADHD at Wave 1 (n = 71; .8%). A comparative group of 9-year olds with ADHD symptoms but no diagnosis was identified based on Strengths & Difficulties Questionnaire (SDQ) hyperactivity/inattention scores, using the procedure recommended by Johnson, Hollis, Marlow, Simms, and Wolke [49]. When using the SDQ to screen for psychopathology, combining SDQ reports from multiple informants produces increased validity and sensitivity [49–52]. Therefore both parent and teacher SDQ data were considered in constructing the Undiagnosed ADHD group. There are numerous methods for aggregating the reports of multiple informants to identify cases of likely disorder [53]; research suggests accepting above-threshold scores in *either* parent or teacher SDQ ratings produces superior sensitivity and specificity than requiring positive screens from *both* informants [49].[2] The SDQ scoring instructions for the hyperactivity/

[2] Johnson et al.'s [49] study of children born preterm found that the more conservative procedure of requiring above-threshold scores in both (rather than either) teacher and parent reports reduced positive ADHD screens in a sample of from 34.2 to 9.6%. Similarly for this study, requiring a score of ≥ 9 in both teacher and parent SDQ reports would have reduced the sample of Undiagnosed ADHD to 81 (.9%)

Table 1 Socio-emotional measures included in the analysis

Measure	Who completed	Wave	Description
Strengths and Difficulties Questionnaire (SDQ) [56]	Parent and teacher	1 and 2	30-item instrument assessing five dimensions of child's behaviour: emotional symptoms, hyperactivity/inattention, conduct problems, peer relationship problems, prosocial behaviour. Higher scores indicate greater pathology on all subscales except prosocial behaviour
Piers-Harris Children's Self-Concept Scale II [57]	Child	1 and 2	60-item instrument assessing five dimensions of the child's self-concept: Behavioural Adjustment, Intellectual & School Status, Physical Appearance & Attributes, Freedom from Anxiety, Popularity, and Happiness & Satisfaction. Higher scores on all subscales indicate more favourable self-concept
Short Mood & Feelings Questionnaire (SMFQ) [58]	Child	2	13-item scale measuring depression in children. Higher scores indicate more depression symptoms

inattention subscale stipulate that values of 9–10 indicate the most extreme cases of 'very high' hyperactivity/inattention. While most studies applying SDQ for screening purposes use a lower threshold [54], the current analysis adopted the conservative criterion of ≥ 9 to ensure the Undiagnosed ADHD group represented individuals who were highly likely to meet diagnostic thresholds. Children who recorded a score of ≥ 9 on either the Wave 1 parent- or teacher-rated hyperactivity/inattention subscale, and whose parent had *not* reported a formal ADHD diagnosis, were identified as the Undiagnosed ADHD group ($n = 582$; 6.8%).

The longitudinal data showed some children who did not report an ADHD diagnosis at age 9 had acquired a diagnosis of an emotional or behavioural disorder by age 13 ($n = 36$; 6.2% of retained Undiagnosed ADHD group). To ensure the analysis provided a clean comparison of diagnosed vs. undiagnosed cases, these children were excluded from the analyses involving Wave 2 outcome data.[3]

For comparative purposes, the remaining sample (who did not report an ADHD diagnosis and were subthreshold on the hyperactivity/inattention subscale) were labelled 'Non-ADHD Controls' ($n = 7915$, 92.4%).

Footnote 2 (continued)

children. This would likely represent an overly conservative estimate of undiagnosed ADHD, given that the accepted prevalence of ADHD in the population is 5% [4] and the rate of existing clinical diagnosis in this sample was .8%.

3 They were not added to the Diagnosed ADHD group due to ambiguities caused by a change in question wording in GUI Wave 2, which queried the presence of "Emotional or behavioural disorders (e.g. ADHD/ADD)" instead of ADHD alone. This wording makes it impossible to establish whether these children held an ADHD diagnosis or another emotional or behavioural diagnosis. To preserve the clarity of the analysis, these ambiguous cases were dropped from any analysis involving Wave 2 data.

Data Analysis

Analysis was conducted using IBM SPSS 24. All analyses using the GUI dataset are advised to incorporate a statistical weight (GROSS) that applies a minimum information loss algorithm to adjust the sample to known population parameters, using census data on social class, socioeconomic status and family structure [48]. This study applied the weights included in the GUI dataset for both waves of data collection. The Wave 2 weights include additional statistical adjustments for participant attrition between study waves (12.2%). Further missing data in the analyses reported below relate to cases where surveys were completed by the parent but not the child (.7% in Wave 1, 1.5% in Wave 2). Such cases were included in analyses involving parent-reported data alone, but excluded from analyses that also involved child-reported measures. Despite the uneven sizes of study groups, power analysis using G*Power [55] indicated the sample had power of .98 to detect a medium-sized group difference ($d = .5$). Pre-tests were performed before all analyses to ensure data met relevant conditions. All hypotheses were two-tailed. A Bonferroni correction was applied to adjust for multiple comparisons, making the criterion for statistical significance $p = .002$. Appropriate measures of effect size were calculated throughout (Cohen's d for t-tests, Cramer's V for Chi square tests, η_p^2 for analyses of covariance).

Descriptive statistics were computed to establish the demographic and clinical characteristics of the study groups. Where appropriate, Chi square or t-tests were used to determine if the Diagnosed and Undiagnosed groups significantly differed in their demographic or clinical characteristics.

Further analysis explored whether the Diagnosed and Undiagnosed ADHD groups differed on three standardised measures of child socio-emotional wellbeing (see Table 1): the SDQ [56], Piers-Harris Children's Self-Concept Scale II [57] and Short Mood & Feelings Questionnaire (SMFQ) [58]. The SMFQ and Piers-Harris were completed by the

Table 2 Socio-demographic profiles of study groups

	Undiagnosed ADHD (n = 582) (%)	Diagnosed ADHD (n = 71) (%)	Non-ADHD Controls (n = 7915) (%)
Gender			
Male	72.1	74.8	49.4
Female	27.9	25.2	50.6
Household structure			
1 Parent, 1–2 children	15.5	17.1	11.1
1 Parent, ≥ 3 children	13.3	18.2	6.1
2 Parents, 1–2 children	31	23.4	35.5
2 Parents, ≥ 3 children	40.2	41.4	47.3
Primary caregiver marital status			
Married and cohabiting	60.3	57.0	78.2
Married and separated	11.5	14.8	6.1
Divorced/widowed	6.3	10.3	2.9
Never married	22.0	17.8	12.7
Household equivalised income quintile			
1st (Lowest)	35.4	26.4	18.8
2nd	17.4	19.5	20.3
3rd	17.4	25.8	20.2
4th	15.2	24.6	20.2
5th (Highest)	14.5	3.7	20.5
Family occupational class			
Professional/managerial	27.7	28.1	42.6
Other non-manual/skilled manual	36.0	32.7	35.4
Semi-skilled/unskilled manual	12.7	11.5	10.9
Not classifiable	23.6	27.7	11.1
Primary caregiver educational attainment			
None/primary	12.8	16.6	5.9
Secondary/vocational	63.1	60.2	60.2
Diploma/certificate	13.3	12.4	16.1
Primary degree	7.8	9.8	11.5
Postgraduate degree	3.1	1.1	6.3
Primary caregiver an Irish citizen			
Yes	92.4	96.1	93.1
No	7.6	3.9	6.9

child. The SDQ was completed by the parent and teacher in Wave 1 but just the parent in Wave 2; since the parent-completed version had more complete longitudinal data, this was selected as the most informative outcome measure. Group differences in the SMFQ were assessed using an ANCOVA test. Since the various subscales of the SDQ and Piers-Harris are conceptually and statistically related, omnibus MANCOVA tests were used for each instrument. For all univariate and multivariate analyses, Wave 1 parent-reported SDQ hyperactivity/inattention scores were included as a covariate, along with other variables that early exploratory tests revealed to significantly differ between study groups (reported below). As group sizes were uneven, Pillai's trace was the most appropriate MANCOVA test statistic. Significant MANCOVA results were followed up with univariate analyses to determine the location of any significant effects.

Results

Do Diagnosed and Undiagnosed Children Differ in Demographic Characteristics?

Table 2 displays the demographic data recorded at Wave 1 for the Undiagnosed ADHD and Diagnosed ADHD group, alongside norms for the Non-ADHD Controls. Comparisons of the Diagnosed and Undiagnosed ADHD groups showed they did not significantly differ in the distribution of gender ($\chi^2[1, n = 653] = .22$, $p = .64$, $V = .02$), two-carer households (χ^2[1, n = 654] = 1.24, p = .25, V = .04), social class (χ^2[2, n = 496] = .18, p = .91, V = .02), parental citizenship (χ^2[1, n = 652] = 1.06, p = .3, V = .04), parental education (χ^2[5, n = 653] = 2.01, p = .85, V = .06) or equivalised income (t[604] = .48, p = .63, d = .07).

Do Diagnosed and Undiagnosed Children Differ in Clinical Characteristics?

General Health

Parents were asked to rate the child's general health in the past year on a scale from 1 (very healthy) to 4 (almost always unwell). In Wave 1, the Diagnosed group (M = 1.65, SD = .59) scored significantly worse than the Undiagnosed group (M = 1.36, SD = .53), t(651) = − 4.38, p < .001, d = .52. However, this difference had disappeared by Wave 2, t(541) = .48, p = .63, d = .06.

At Wave 1, 12.7% of the Diagnosed group had made more than 4 visits to the GP in the previous year, compared to 6.5% of the Undiagnosed and 5.8% of the Non-ADHD group. In Wave 2, 11.9% of Diagnosed, 4.4% of Undiagnosed and 3.8% Non-ADHD children reported > 4 GP contacts. The difference between Diagnosed and Undiagnosed groups fell short of Bonferroni-corrected statistical significance at both Wave 1 (χ^2[1, n = 653] = 3.58, p = .06, V = .07) and Wave 2 (χ^2[1, n = 542] = 6.8, p = .01, V = .02).

At Wave 1, there was no significant difference in the proportion of children who had spent numerous (> 2) nights in hospital over their lifetime, χ^2(1, n = 654) = .90, p = .34, V = .04. However, by Wave 2, Diagnosed children were significantly more likely to have spent > 2 nights in hospital (56.7%) compared to Undiagnosed children (33.5%), χ^2(1, n = 544) = 13.63, p < .001, V = .16.

Of the 9-year olds who held a clinical diagnosis of ADHD, 13.4% had been diagnosed in the last 6 months, 14.1% 6–12 months ago, 19.7% 1–2 years ago and 52.8% over 2 years ago. The Diagnosed group scored significantly higher on parent-rated hyperactivity/inattention than the Undiagnosed group at Wave 1 (t[651] = − 4.09, p < .001, d = .59) and Wave 2 (t[541] = − 6.41, p < .001, d = .94). The groups did not differ on teacher-rated hyperactivity/inattention, t(644) = − .54, p = .59, d = .08.

Wave 1 collected some limited data on children's engagement with mental health services. In a series of questions about contact with health professionals, parents were asked how frequently in the past year they had discussed their child's health with a professional such as a psychologist, psychiatrist or counsellor. Half (50%) of the Diagnosed group reported multiple (> 1) contacts, compared with 7.1% of the Undiagnosed and 2% of the Non-ADHD sample. The difference between the Diagnosed and Undiagnosed group was statistically significant, χ^2(1, n = 650) = 112.87, p < .001, V = .42. Responses to the same question in Wave 2 showed that 52.6% of the Diagnosed group had multiple consultations, compared with 5.8% of the Undiagnosed ADHD group and 2.3% of the Non-ADHD sample. The difference between the Diagnosed and Undiagnosed group was again significant, χ^2(1, n = 542) = 125.73, p < .001, V = .48.

Wave 2 collected some additional information regarding service-use, which was absent from Wave 1. Of the Diagnosed group, 25.3% reported receiving support from a psychiatrist, 12.4% a school psychologist and 27.8% an out-of-school psychologist. Equivalent figures from the Undiagnosed ADHD group show 0.5% engaging with a psychiatrist, 6% a school psychologist and 2.5% an out-of-school psychologist. Parents who reported their child had a mental health condition or disability were also asked whether the child had been prescribed any medication for emotional or behavioural problems. 41.1% of the Diagnosed ADHD group reported a prescription.[4] No information on engagement with psychotherapeutic interventions was available.

Cognitive Ability

Standardised psychometric tests, the Drumcondra Reasoning Tests, assessed children's Verbal Reasoning and Numerical Ability. Both scales were converted into logit scores that adjusted the percentage correct score using two parameters (difficulty and discrimination) for each item. In Wave 1, the Diagnosed group scored significantly poorer than the Undiagnosed group on numerical reasoning (t[624] = 3.25, p = .001, d = .45), but the difference on verbal reasoning fell short of Bonferroni-corrected statistical significance (t[616] = 2.39, p = .017, d = .32). By Wave 2 there was no significant difference in either verbal (t[461] = .865, p = .39, d = .15) or numerical (t[448] = 1.94, p = .053, d = .31) performance.

Child Wellbeing

Table 3 shows the unadjusted means for the Diagnosed, Undiagnosed and non-ADHD groups on socio-emotional variables (parent-rated SDQ, Piers-Harris, SMFQ) for both study waves.

Four MANCOVAs were performed to establish whether the Diagnosed and Undiagnosed ADHD groups differed on socio-emotional measures from both waves. MANCOVAs controlled for five variables that the above analyses showed to differ between study groups: Wave 1 parent-reported

[4] Unfortunately, since questions about medication were not asked of parents who did not report a mental health condition or disability, equivalent information is not available for the Undiagnosed ADHD group.

Table 3 Mean scores on socio-emotional variables

	Undiagnosed ADHD (n = 582)		Diagnosed ADHD (n = 71)		Non-ADHD controls (n = 7915)	
	M	*SD*	*M*	*SD*	*M*	*SD*
Wave 1						
Piers-Harris						
Behavioural adjustment	10.14	3.15	10.11	3.44	11.80	2.36
Intellectual and school status	10.98	3.31	10.84	3.42	12.43	2.79
Physical appearance and attributes	7.25	2.64	7.40	2.62	7.61	2.27
Freedom from anxiety	9.51	3.27	9.68	3.04	10.83	2.84
Popularity	7.41	2.69	7.43	2.48	8.68	2.38
Happiness and satisfaction	7.88	2.02	7.81	1.84	8.74	1.63
Total self-concept	41.43	10.23	41.64	10.08	46.69	8.42
SDQ						
Emotional	2.90	2.45	4.37	2.52	2.06	1.99
Conduct	2.63	2.01	4.10	2.10	1.26	1.39
Hyperactivity/inattention	7.03	2.76	8.40	1.78	2.90	2.18
Peer relationships	2.12	2.05	3.69	1.99	1.17	1.40
Total problems	14.70	6.37	20.56	5.76	7.38	4.71
Prosocial behaviour	8.45	1.86	7.42	2.09	8.92	1.40
Wave 2						
Piers-Harris						
Behavioural adjustment	11.64	2.46	10.14	3.32	12.55	2.13
Intellectual and school status	11.29	3.22	10.87	3.11	11.98	3.18
Physical appearance and attributes	8.01	2.45	8.25	2.21	7.81	2.42
Freedom from anxiety	11.04	2.64	9.57	4.12	10.74	3.01
Popularity	9.38	2.32	8.21	2.81	9.77	2.16
Happiness and satisfaction	8.64	1.70	8.84	1.31	8.58	1.69
Total self-concept	46.46	8.51	42.67	9.85	47.72	8.43
SDQ						
Emotional	2.47	2.36	4.65	2.54	1.84	1.96
Conduct	2.22	2.03	3.16	2.03	1.14	1.38
Hyperactivity/inattention	5.71	2.66	7.87	1.91	2.57	2.28
Peer relationships	1.77	1.82	4.08	2.36	1.07	1.42
Total problems	12.17	6.38	19.77	6.72	6.62	4.97
Prosocial behaviour	8.26	1.77	7.51	1.99	8.86	1.49
SMFQ						
Total depression	4.45	4.78	7.62	7.77	3.78	4.30

hyperactivity/inattention, Wave 1 general health, Wave 1 verbal and numerical performance, and having multiple contacts with mental health professionals in the previous year. The latter variable can be conceptualised as a proxy for mental health service engagement and had two versions corresponding to each study wave, with analyses using the version from the wave commensurate to the outcome variables under consideration.

Wave 1

Using Pillai's trace, the MANCOVA for Wave 1 parent-rated SDQ scores showed that (controlling for hyperactivity/inattention symptoms, general health, cognitive ability and service engagement), the Diagnosed and Undiagnosed ADHD groups did not significantly differ on SDQ scores, $V = .02$, $F(4, 600) = 2.33$, $p = .06$, $\eta_p^2 = .02$. Similarly, a MANCOVA for Wave 1 Piers-Harris scores showed no significant difference between study groups' self-concepts, $V = .01$, $F(6, 544) = 1.06$, $p = .38$, $\eta_p^2 = .01$.

Wave 2

Pillai's trace showed that (controlling for hyperactivity/inattention symptoms, general health, cognitive ability and service engagement) the groups significantly differed on Wave 2 parent-rated SDQ scores, $V = .06$, $F(4, 493) = 7.6$, $p < .001$, $\eta_p^2 = .06$. Follow-up univariate ANCOVAs on the four SDQ subscales confirmed that the Diagnosed group showed poorer scores on the Emotional ($F[1, 496] = 7.67$, $p = .01$, $\eta_p^2 = .02$), Peer Relationships ($F[1, 496] = 28.61$, $p < .001$, $\eta_p^2 = .06$) and Prosocial Behaviour ($F[1, 496] = 5.42$, $p = .02$, $\eta_p^2 = .01$) scales. There was no significant difference on Conduct ($F[1, 496] = .34$, $p = .56$, $\eta_p^2 = .001$).

Pillai's trace showed the groups significantly differed on Wave 2 Piers-Harris scores, $V = .05$, $F(6, 471) = 4.19$, $p < .001$, $\eta_p^2 = .05$. Separate univariate tests indicated the Undiagnosed group had more favourable self-concept in the domains of Behavioural Adjustment ($F[1, 476] = 8.2$, $p = .004$, $\eta_p^2 = .02$), Popularity ($F[1, 476] = 6.35$, $p = .01$, $\eta_p^2 = .01$) and Freedom from Anxiety ($F[1, 476] = 4.28$, $p = .04$, $\eta_p^2 = .01$). The groups did not differ on Physical Appearance & Attributes ($F[1, 476] = .12$, $p = .74$, $\eta_p^2 = .00$), Intellectual & School Status ($F[1, 476] = .09$, $p = .77$, $\eta_p^2 = .0$) or Happiness & Satisfaction ($F[1, 476] = 1.09$, $p = .3$, $\eta_p^2 = .002$).

A one-way ANCOVA explored whether the groups differed on the SMFQ, controlling for hyperactivity/inattention, general health, cognitive ability and service engagement. Although the adjusted means showed the Diagnosed group had more depression symptoms, this difference fell short of statistical significance, $F(1, 483) = 3.55$, $p = .06$, $\eta_p^2 = .01$.

Discussion

The current study represents the first to use a longitudinal community sample to compare the demographic, clinical and socio-emotional characteristics of children with hyperactivity/inattention symptoms, who have and have not received a diagnosis of ADHD. The analysis revealed no significant differences in the socio-demographic profiles of 9-year olds with hyperactivity/inattention who had and who had not received a diagnosis of ADHD. Nine-year olds with a diagnosis had poorer general health and cognitive ability than their undiagnosed hyperactive peers, but these differences had disappeared by the time the children turned 13. The opposite developmental pattern applied to indicators of socio-emotional wellbeing. Controlling for the level of hyperactivity/inattention, health, cognitive ability and service engagement, diagnosed and undiagnosed 9-year olds did not significantly differ in their socio-emotional wellbeing. However, 4 years later the two groups had diverged, with multivariate analyses showing the diagnosed group as more impaired on both the SDQ and Piers-Harris. Specifically, relative to undiagnosed hyperactive children, 9-year olds who held an ADHD diagnosis had, by age 13, more emotional and peer relationship problems, worse prosocial behaviour, and poorer self-concept in dimensions of behavioural adjustment, popularity and anxiety. Diagnosed children also showed more depression symptoms, although this difference did not reach statistical significance.

The relation of ADHD diagnosis with longitudinal socio-emotional disadvantage in children with hyperactivity/inattention is a novel finding in the literature [47] and requires explanation. One potential explanation is that children whose hyperactivity/inattention was accompanied by more severe socio-emotional impairment, whether as comorbid disorders or subclinical distress, were more likely to come to clinical attention and receive a diagnosis by age 9. According to this explanation, even if the diagnosis and any clinical interventions that followed had positive effects, such improvements were not sufficient to rectify the baseline level of socio-emotional impairment. The GUI dataset does not contain detailed information on specific mental health diagnoses aside from ADHD, leaving it unclear whether (for instance) comorbid mood or anxiety disorders could impact the trajectories observed. Descriptive statistics do show diagnosed 9-year olds scored higher on all SDQ problem subscales than the undiagnosed ADHD group, whose difficulties were more localised to hyperactivity/inattention. However, analysis indicated that once the extent of hyperactivity/inattention and other relevant variables were controlled, group differences in Wave 1 SDQ or Piers-Harris scores were not statistically significant. Rather, differences between diagnosed and undiagnosed children arose in the intervening years before the children reached age 13.

A further explanation is that children who received a diagnosis by 9 years were distinct on some other variables, that over time compromised socio-emotional development. No socio-demographic differences were apparent between the diagnosed and undiagnosed cohorts, but detailed information on child ethnicity, which is known to influence patterns of diagnosis [1, 3], was lacking. It is possible that a familial factor not included in the analysis mutually affected the likelihood of parents seeking/clinicians giving a diagnosis, and the development of socio-emotional problems. For instance, parental beliefs about the nature, aetiology and severity of a child's hyperactive behaviour influence service contact and pathways [45]; these same variables may modulate socio-emotional outcomes (e.g. parents' perception of their child's behaviour as burdensome could both increase the likelihood of diagnosis and threaten the child's self-esteem). The presence of ADHD in parents or siblings, which is not queried in the GUI data, could also mutually affect the likelihood of diagnosis and socio-emotional outcomes. Similarly, other medical, cognitive or behavioural difficulties could encourage both diagnosis and longitudinal socio-emotional problems. Diagnosed children had significantly worse general health, hyperactivity/inattention symptoms and cognitive ability at Wave 1, which likely increased the probability of the child attracting clinical attention. It is unlikely these early disadvantages directly caused the later socio-emotional difficulties, as cognitive and health differences had disappeared by Wave 2 and the MANOVA analyses controlled for individual variation in these variables. Nevertheless, it remains possible that early cognitive or behavioural difficulties could exert indirect effects on children's life trajectories that contribute to suboptimal socio-emotional development, for example by affecting school placements or parenting practices. The timespan of 9–13 years would incorporate most children's move to secondary school; perhaps this transition poses particular challenges for children with prior experience of cognitive or behavioural difficulties. Further longitudinal research is required to clarify the chains of ripple-effects that early medical, cognitive or behavioural disadvantage may trigger.

A further explanation for the diagnosed group's socio-emotional disadvantage is that receiving the diagnosis altered the child's life trajectory in a manner disadvantageous for socio-emotional functioning. Previous literature suggests numerous ways an ADHD diagnosis can negatively affect a young person's life. A diagnosis can represent a threat to a young person's self-concept [29, 30, 32, 34, 35] and mark them as 'different' from their peers [30, 32, 34, 36]. Holding an ADHD diagnosis can prompt negative responses from peers [59–65] and expose a young person to stigma and bullying [30, 32–34, 36]. An ADHD diagnosis can also lead to lowered aspirations and expectations from teachers and parents [59, 64, 66]. Such perceptions could be internalised by the child, contributing to the lowered self-worth identified by the current study, or produce changes in parenting styles that contribute to suboptimal socio-emotional development. Although previous research confirms diagnosis can entail many benefits for a young person's self-concept and social relations [29–34, 36], these were not evident in the current data. While the diagnosed group were statistically equivalent to the undiagnosed group on some dimensions of socio-emotional wellbeing, such as conduct problems and intellectual self-esteem, on no dimension were the diagnosed group advantaged relative to the undiagnosed group.

It is possible that any negative socio-emotional effects of diagnosis could be moderated by effective treatments that diagnosis may facilitate. In Harpin et al.'s [43] meta-analysis of self-esteem and social outcomes, treated ADHD generally showed more positive outcomes than untreated ADHD. It is difficult to partial out the effects of diagnosis and treatment in the current analysis, since GUI contains relatively sparse information on service use. Half of the diagnosed group reported multiple recent contacts with a mental health professional and 41.1% had been prescribed a psychotropic medication by 13 years old. This is somewhat lower than US medication rates of approximately 60% [3, 12]. The diagnosed group's negative outcomes may be particularly driven by the half who were apparently disengaged from clinical services. It could equally be argued that the resilience of the undiagnosed ADHD group might be due to their accessing effective therapeutic support outside clinical contexts: a British study of children with ADHD features found that if parents defined the primary problem as emotional or behavioural rather than hyperactivity, they were more likely to access educational services for behavioural or emotional problems rather than primary or specialist care [45]. However, in the current study use of educational or community mental health supports among the undiagnosed group was low (< 6%), making it unlikely their superior socio-emotional outcomes resulted from non-clinical interventions.[5] The MANOVA analyses did control for service engagement (having multiple contacts with mental health professionals in the previous year, which presumably would be necessary if the child was actively receiving treatment). Results suggested diagnosis had significant negative associations with longitudinal socio-emotional outcomes, independent of whether children were engaging with services. However, given the crude nature of the service engagement variable, it is possible the analysis did not fully capture variation related to the receipt of sustained, evidence-based treatment. More nuanced data on service use is necessary to fully explore the interactions between the socio-emotional effects of diagnosis and treatment pathways.

A limitation of this study was the relatively small number of cases of ADHD diagnoses available for analysis ($n = 71$, .8%). This may have compromised the analysis' power to detect differences between study groups at Wave 1. The low prevalence of parent-reported ADHD diagnoses is consistent with other European community studies [52, 67, 68], which tend to show lower figures than US-based research [2]. Parental under-reporting may be a factor in the low prevalence of recorded diagnosis; future research should incorporate corroborating evidence on diagnosis from clinical settings. It should be noted that GUI coverage was high, with the Wave 1 sample including approximately one in seven of all 9-year-olds nationally [48]. It is possible that families with children with behavioural dysfunctions were less likely to opt into the study; although as such dynamics likely apply to both the diagnosed and undiagnosed ADHD cohorts, the between-group comparison remains valid. Nevertheless, the rate of ADHD diagnosis revealed by the current study (.8%) is well below the estimated prevalence of 5% [4], despite the level of above-threshold hyperactivity/

[5] GUI did not ask those without a diagnosis whether their child was taking psychotropic medication, so it is not possible to determine whether some undiagnosed children were nevertheless undergoing pharmaceutical intervention. The lack of this data for the undiagnosed group also means it was not possible to include pharmaceutical treatment as a covariate in the analysis.

inattention (6.8%) being similar to other community studies [69]. The lack of socio-demographic biases in the distribution of diagnoses is internationally unusual [3, 10, 11], but may reflect limitations of statistical power given the small sample of diagnosed children. The low prevalence of diagnosis may reflect a culturally conservative approach to childhood psychiatric diagnosis in Ireland, as appears to be the case in the UK [2]. It is worth noting that the .8% prevalence figure is based on data collected in 2007–2008 and may have increased in the intervening decade. GUI is also collecting data on a cohort of children 8–9 years younger than those in the current study, which may enlighten historical changes in rates of diagnosis.

The low prevalence of ADHD diagnosis may contribute to the negative socio-emotional trajectories observed in this study. Infrequency of diagnosis may both reflect and reinforce a cultural context where psychiatric diagnoses in childhood are highly stigmatised. For instance, a substantial minority of Irish GPs believe that an ADHD diagnosis is stigmatising and that parents seek diagnoses to excuse their child's bad behaviour [27], which may prevent them from referring children for CAMHS assessment. Growing up with a diagnosis in a society where very few others share this classification may be particularly difficult, which could contribute to the suboptimal socio-emotional trajectories suggested by this analysis. This highlights the importance of cultural context in mediating the socio-emotional repercussions of receiving a psychiatric diagnosis. The patterns observed in this study may not generalise to cultures in which ADHD diagnoses are more common and accepted. Future research should investigate whether the findings of this Irish study hold in other international contexts.

The possibility that social stigma is a key influence on the negative socio-emotional effects of diagnosis reinforces the importance of developing effective stigma reduction strategies. It is notable that diagnosed children in this study ranked worse than undiagnosed children in parent-reported peer relationships and self-reported popularity at age 13. There is an established correlation between ADHD and peer problems [70–72], which has usually been explained as ensuing from ADHD's core symptoms and their interpersonal corollaries [73]. The current research indicates diagnosed children have poorer peer relations than undiagnosed children, even when the extent of hyperactivity/inattention symptoms is controlled. This may suggest the diagnostic label and attendant stigma contribute to peer exclusion or victimisation in early adolescence. While most western countries, including Ireland, consistently run mental health stigma reduction campaigns, these are usually targeted at adult populations and rarely informed by evidenced attitude and behaviour change techniques [74, 75]. Developing tailored stigma reduction programmes for child and young adult populations should be a priority for those interested in improving the socio-emotional trajectories of young people with psychiatric diagnoses.

However, cultural change unfortunately takes time, and clinicians must work to optimise a child's outcomes within the cultural context in which they currently live. The current study suggests that clinicians should be mindful of the potentially negative socio-emotional processes that may accompany an ADHD diagnosis. Some experts have proposed a system of 'stepped diagnosis', where formal diagnosis is deferred until after a 'watchful waiting' and minimal intervention phase have observed no symptomatic change [8, 9]. This may be appropriate in contexts where an ADHD diagnosis is stigmatised and a diagnosis is not a precondition for initiating interventions. The longitudinal socio-emotional disadvantage attached to an ADHD diagnosis in this study also highlights the importance of considering a child's trajectory holistically, rather than solely in terms of symptom reduction.

Summary

Most debates about ADHD under- or over-diagnosis have focused on risks pertaining to treatment neglect or inappropriate medication [2, 6, 9]; there has been relatively little consideration of diagnosis' risks and benefits for the child's developing self-concept, social relations and emotional wellbeing. The current research suggests that all else being equal, children with diagnosed ADHD fare worse on such dimensions than their peers with similar symptomatology but no diagnosis. Further research is required to disentangle whether this association is due to socio-emotional pathways triggered by the diagnosis itself, or other confounding factors in children's biological, familial, or educational backgrounds.

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Compliance with Ethical Standards

Conflicts of interest The authors declare that they have no conflicts of interest.

Ethical Approval The GUI study was conducted in accordance with the 1964 Helsinki Declaration and its later amendments and received ethical approval from the Irish Health Research Board's Research Ethics Committee. Informed consent was obtained from all participants.

References

1. Schneider H, Eisenberg D (2006) Who receives a diagnosis of attention-deficit/hyperactivity disorder in the United States

elementary school population? Pediatrics 117:e601–e609. https://doi.org/10.1542/peds.2005-1308
2. Taylor E (2017) Attention deficit hyperactivity disorder: overdiagnosed or diagnoses missed? Arch Dis Child 102:376–379. https://doi.org/10.1136/archdischild-2016-310487
3. Visser SN, Danielson ML, Bitsko RH et al (2014) Trends in the parent-report of health care provider-diagnosed and medicated attention-deficit/hyperactivity disorder: United States, 2003-2011. J Am Acad Child Adolesc Psychiatry 53:34–46.e2. https://doi.org/10.1016/j.jaac.2013.09.001
4. Polanczyk GV, Willcutt EG, Salum GA et al (2014) ADHD prevalence estimates across three decades: an updated systematic review and meta-regression analysis. Int J Epidemiol. https://doi.org/10.1093/ije/dyt261
5. Ginsberg Y, Quintero J, Anand E et al (2014) Underdiagnosis of attention-deficit/hyperactivity disorder in adult patients: a review of the literature. Prim Care Companion CNS Disord. https://doi.org/10.4088/PCC.13r01600
6. Hamed AM, Kauer AJ, Stevens HE (2015) Why the diagnosis of attention deficit hyperactivity disorder matters. Front Psychiatry. https://doi.org/10.3389/fpsyt.2015.00168
7. Sayal K, Prasad V, Daley D et al (2018) ADHD in children and young people: prevalence, care pathways, and service provision. Lancet Psychiatry 5:175–186. https://doi.org/10.1016/S2215-0366(17)30167-0
8. Batstra L, Frances A (2012) Holding the line against diagnostic inflation in psychiatry. PPS 81:5–10. https://doi.org/10.1159/000331565
9. Thomas R, Mitchell GK, Batstra L (2013) Attention-deficit/hyperactivity disorder: are we helping or harming? BMJ 347:f6172. https://doi.org/10.1136/bmj.f6172
10. Akinbami LJ, Liu X, Pastor PN, Reuben CA (2011) Attention deficit hyperactivity disorder among children aged 5–17 years in the United States, 1998–2009. National Center for Health Statistics, Hyattsville, MD
11. Boyle CA, Boulet S, Schieve LA et al (2011) Trends in the prevalence of developmental disabilities in US children, 1997-2008. Pediatrics 127:1034–1042. https://doi.org/10.1542/peds.2010-2989
12. Danielson ML, Bitsko RH, Ghandour RM et al (2018) Prevalence of parent-reported ADHD diagnosis and associated treatment among U.S. children and adolescents, 2016. J Clin Child Adolesc Psychol 47:199–212. https://doi.org/10.1080/15374416.2017.1417860
13. Conrad P, Bergey MR (2014) The impending globalization of ADHD: notes on the expansion and growth of a medicalized disorder. Soc Sci Med 122:31–43. https://doi.org/10.1016/j.socscimed.2014.10.019
14. Scheffler RM, Hinshaw SP, Modrek S, Levine P (2007) The global market for ADHD medications. Health Aff (Millwood) 26:450–457. https://doi.org/10.1377/hlthaff.26.2.450
15. Bruchmüller K, Margraf J, Schneider S (2012) Is ADHD diagnosed in accord with diagnostic criteria? Overdiagnosis and influence of client gender on diagnosis. J Consult Clin Psychol 80:128–138. https://doi.org/10.1037/a0026582
16. Quinn PO, Madhoo M (2014) A review of attention-deficit/hyperactivity disorder in women and girls: uncovering this hidden diagnosis. Prim Care Companion CNS Disord. https://doi.org/10.4088/PCC.13r01596
17. Chen M-H, Lan W-H, Bai Y-M et al (2016) Influence of relative age on diagnosis and treatment of attention-deficit hyperactivity disorder in Taiwanese children. J Pediatr 172:162–167.e1. https://doi.org/10.1016/j.jpeds.2016.02.012
18. Elder TE (2010) The importance of relative standards in ADHD diagnoses: evidence based on exact birth dates. J Health Econ 29:641–656. https://doi.org/10.1016/j.jhealeco.2010.06.003
19. Evans WN, Morrill MS, Parente ST (2010) Measuring inappropriate medical diagnosis and treatment in survey data: the case of ADHD among school-age children. J Health Econ 29:657–673. https://doi.org/10.1016/j.jhealeco.2010.07.005
20. Halldner L, Tillander A, Lundholm C et al (2014) Relative immaturity and ADHD: findings from nationwide registers, parent- and self-reports. J Child Psychol Psychiatry 55:897–904. https://doi.org/10.1111/jcpp.12229
21. Hoshen MB, Benis A, Keyes KM, Zoëga H (2016) Stimulant use for ADHD and relative age in class among children in Israel. Pharmacoepidemiol Drug Saf 25:652–660. https://doi.org/10.1002/pds.3962
22. Morrow RL, Garland EJ, Wright JM et al (2012) Influence of relative age on diagnosis and treatment of attention-deficit/hyperactivity disorder in children. CMAJ 184:755–762. https://doi.org/10.1503/cmaj.111619
23. Sayal K, Chudal R, Hinkka-Yli-Salomäki S et al (2017) Relative age within the school year and diagnosis of attention-deficit hyperactivity disorder: a nationwide population-based study. The Lancet Psychiatry 4:868–875. https://doi.org/10.1016/S2215-0366(17)30394-2
24. Whitely M, Lester L, Phillimore J, Robinson S (2017) Influence of birth month on the probability of Western Australian children being treated for ADHD. Med J Aust 206:85. https://doi.org/10.5694/mja16.00398
25. Zoëga H, Valdimarsdóttir UA, Hernández-Díaz S (2012) Age, academic performance, and stimulant prescribing for ADHD: a nationwide cohort study. Pediatrics 130:1012–1018. https://doi.org/10.1542/peds.2012-0689
26. Health Service Exeecutive (2014) Fifth annual child & adolescent mental health service report. Health Service Exeecutive, Dublin
27. Adamis D, Tatlow-Golden M, Gavin B, McNicholas F (2018) General practitioners' (GP) attitudes and knowledge about attention deficit hyperactivity disorder (ADHD) in Ireland. Ir J Med Sci. https://doi.org/10.1007/s11845-018-1804-3
28. O'Connor C, Kadianaki I, Maunder K, McNicholas F (2018) How does psychiatric diagnosis affect young people's sense of self and social identity? A systematic review and synthesis of the qualitative literature. Soc Sci Med 212:94–119. https://doi.org/10.1016/j.socscimed.2018.07.011
29. Cheung KKW, Wong ICK, Ip P et al (2015) Experiences of adolescents and young adults with ADHD in Hong Kong: treatment services and clinical management. BMC Psychiatry. https://doi.org/10.1186/s12888-015-0478-x
30. Cooper P, Shea T (1998) Pupils' perceptions of AD/HD. Emot Behav Diffic 3:36–48. https://doi.org/10.1080/1363275980030306
31. Honkasilta J, Vehmas S, Vehkakoski T (2016) Self-pathologizing, self-condemning, self-liberating: youths' accounts of their ADHD-related behavior. Soc Sci Med 150:248–255. https://doi.org/10.1016/j.socscimed.2015.12.030
32. Kendall J, Hatton D, Beckett A, Leo M (2003) Children's accounts of attention-deficit/hyperactivity disorder. Adv Nurs Sci 26:114–130
33. Singh I (2011) A disorder of anger and aggression: children's perspectives on attention deficit/hyperactivity disorder in the UK. Soc Sci Med 73:889–896. https://doi.org/10.1016/j.socscimed.2011.03.049
34. Travell C, Visser J (2006) 'ADHD does bad stuff to you': young people's and parents' experiences and perceptions of attention deficit hyperactivity disorder (ADHD). Emot Behav Diffic 11:205–216. https://doi.org/10.1080/13632750600833924
35. Krueger M, Kendall J (2001) Descriptions of self: an exploratory study of adolescents with ADHD. J Child Adolesc Psychiatr Nurs 14:61–72. https://doi.org/10.1111/j.1744-6171.2001.tb00294.x
36. Hallberg U, Klingberg G, Setsaa W, Moller A (2010) Hiding parts of one's self from others—a grounded theory study on teenagers

diagnosed with ADHD. Scand J Disabil Res 12:211–220. https://doi.org/10.1080/15017410903478964
37. MTA Cooperative Group (2004) National Institute of Mental Health Multimodal Treatment Study of ADHD follow-up: changes in effectiveness and growth after the end of treatment. Pediatrics 113:762–769
38. Shaw M, Hodgkins P, Caci H et al (2012) A systematic review and analysis of long-term outcomes in attention deficit hyperactivity disorder: effects of treatment and non-treatment. BMC Med 10:99. https://doi.org/10.1186/1741-7015-10-99
39. Storebø OJ, Ramstad E, Krogh HB et al (2015) Methylphenidate for children and adolescents with attention deficit hyperactivity disorder (ADHD). Cochrane Database Syst Rev. https://doi.org/10.1002/14651858.CD009885.pub2
40. Sonuga-Barke EJS, Brandeis D, Cortese S et al (2013) Nonpharmacological interventions for ADHD: systematic review and meta-analyses of randomized controlled trials of dietary and psychological treatments. Am J Psychiatry 170:275–289. https://doi.org/10.1176/appi.ajp.2012.12070991
41. Zwi M, Jones H, Thorgaard C et al (2011) Parent training interventions for attention deficit hyperactivity disorder (ADHD) in children aged 5 to 18 years. Cochrane Database Syst Rev. https://doi.org/10.1002/14651858.CD003018.pub3
42. Arnold LE, Hodgkins P, Caci H et al (2015) Effect of treatment modality on long-term outcomes in attention-deficit/hyperactivity disorder: a systematic review. PLoS ONE 10:e0116407. https://doi.org/10.1371/journal.pone.0116407
43. Harpin V, Mazzone L, Raynaud JP et al (2016) Long-term outcomes of ADHD: a systematic review of self-esteem and social function. J Atten Disord 20:295–305. https://doi.org/10.1177/1087054713486516
44. Fischer M, Barkley RA, Fletcher KE, Smallish L (1993) The adolescent outcome of hyperactive children: predictors of psychiatric, academic, social, and emotional adjustment. J Am Acad Child Adolesc Psychiatry 32:324–332. https://doi.org/10.1097/00004583-199303000-00013
45. Sayal K, Goodman R, Ford T (2006) Barriers to the identification of children with attention deficit/hyperactivity disorder. J Child Psychol Psychiatry 47:744–750. https://doi.org/10.1111/j.1469-7610.2005.01553.x
46. Rey JM, Sawyer MG (2003) Are psychostimulant drugs being used appropriately to treat child and adolescent disorders? Br J Psychiatry 182:284–286. https://doi.org/10.1192/bjp.182.4.284
47. Able SL, Johnston JA, Adler LA, Swindle RW (2007) Functional and psychosocial impairment in adults with undiagnosed ADHD. Psychol Med 37:97–107. https://doi.org/10.1017/S0033291706008713
48. Thornton M, Williams J, McCrory C et al (2016) Growing up in Ireland National Longitudinal Study of children: design, instrumentation and procedures for the child cohort at wave two (13 years). Department of Children and Youth Affairs, Dublin
49. Johnson S, Hollis C, Marlow N et al (2014) Screening for childhood mental health disorders using the strengths and difficulties questionnaire: the validity of multi-informant reports. Dev Med Child Neurol 56:453–459. https://doi.org/10.1111/dmcn.12360
50. Goodman R, Ford T, Simmons H et al (2000) Using the strengths and difficulties questionnaire (SDQ) to screen for child psychiatric disorders in a community sample. Br J Psychiatry 177:534–539
51. Goodman R, Renfrew D, Mullick M (2000) Predicting type of psychiatric disorder from strengths and difficulties questionnaire (SDQ) scores in child mental health clinics in London and Dhaka. Eur Child Adolesc Psychiatry 9:129–134. https://doi.org/10.1007/s007870050008
52. Russell G, Rodgers LR, Ford T (2013) The strengths and difficulties questionnaire as a predictor of parent-reported diagnosis of autism spectrum disorder and attention deficit hyperactivity disorder. PLoS ONE 8:e80247. https://doi.org/10.1371/journal.pone.0080247
53. van Dulmen MHM, Egeland B (2011) Analyzing multiple informant data on child and adolescent behavior problems: predictive validity and comparison of aggregation procedures. Int J Behav Dev 35:84–92. https://doi.org/10.1177/0165025410392112
54. Silva TBF, Osório FL, Loureiro SR (2015) SDQ: discriminative validity and diagnostic potential. Front Psychol. https://doi.org/10.3389/fpsyg.2015.00811
55. Faul F, Erdfelder E, Lang A-G, Buchner A (2007) G*Power 3: a flexible statistical power analysis program for the social, behavioral, and biomedical sciences. Behav Res Methods 39:175–191
56. Goodman R (1997) The strengths and difficulties questionnaire: a research note. J Child Psychol Psychiatry 38:581–586. https://doi.org/10.1111/j.1469-7610.1997.tb01545.x
57. Piers EV, Herzberg DS (2007) Piers-Harris children's self-concept scale–manual, 2nd edn. Western Psychological Services, Los Angeles, CA
58. Angold A, Costello EJ, Messer SC, Pickles A (1995) Development of a short questionnaire for use in epidemiological studies of depression in children and adolescents. Int J Methods Psychiatr Res 5:237–249
59. Batzle CS, Weyandt LL, Janusis GM, DeVietti TL (2010) Potential impact of ADHD with stimulant medication label on teacher expectations. J Atten Disord 14:157–166. https://doi.org/10.1177/1087054709347178
60. Canu WH, Newman ML, Morrow TL, Pope DLW (2008) Social appraisal of adult ADHD stigma and influences of the beholder's big five personality traits. J Atten Disord 11:700–710. https://doi.org/10.1177/1087054707305090
61. Chew BL, Jensen SA, Rosén LA (2009) College students' attitudes toward their ADHD peers. J Atten Disord 13:271–276. https://doi.org/10.1177/1087054709333347
62. Harris MJ, Milich R, Corbitt EM et al (1992) Self-fulfilling effects of stigmatizing information on children's social interactions. J Pers Soc Psychol 63:41–50. https://doi.org/10.1037/0022-3514.63.1.41
63. O'Driscoll C, Heary C, Hennessy E, McKeague L (2012) Explicit and implicit stigma towards peers with mental health problems in childhood and adolescence. J Child Psychol Psychiatry 53:1054–1062. https://doi.org/10.1111/j.1469-7610.2012.02580.x
64. Ohan JL, Visser TAW, Moss RG, Allen NB (2013) Parents' stigmatizing attitudes toward psychiatric labels for ADHD and depression. Psychiatr Serv 64:1270–1273. https://doi.org/10.1176/appi.ps.201200578
65. Walker JS, Coleman D, Lee J et al (2008) Children's stigmatization of childhood depression and ADHD: magnitude and demographic variation in a national sample. J Am Acad Child Adolesc Psychiatry 47:912–920. https://doi.org/10.1097/CHI.0b013e318179961a
66. Ohan JL, Visser TAW, Strain MC, Allen L (2011) Teachers' and education students' perceptions of and reactions to children with and without the diagnostic label "ADHD". J Sch Psychol 49:81–105. https://doi.org/10.1016/j.jsp.2010.10.001
67. Ford T, Goodman R, Meltzer H (2003) The British child and adolescent mental health survey 1999: the prevalence of DSM-IV disorders. J Am Acad Child Adolesc Psychiatry 42:1203–1211. https://doi.org/10.1097/00004583-200310000-00011
68. Kvist AP, Nielsen HS, Simonsen M (2013) The importance of children's ADHD for parents' relationship stability and labor supply. Soc Sci Med 88:30–38. https://doi.org/10.1016/j.socscimed.2013.04.001
69. Meltzer H, Gatward R, Goodman R, Ford T (2000) Mental health of children and adolescents in Great Britain. The Stationery Office, London
70. Bagwell CL, Molina BSG, Pelham WE, Hoza B (2001) Attention-deficit hyperactivity disorder and problems in peer relations:

predictions from childhood to adolescence. J Am Acad Child Adolesc Psychiatry 40:1285–1292. https://doi.org/10.1097/00004583-200111000-00008
71. Barkley RA, Fischer M, Smallish L, Fletcher K (2006) Young adult outcome of hyperactive children: adaptive functioning in major life activities. J Am Acad Child Adolesc Psychiatry 45:192–202. https://doi.org/10.1097/01.chi.0000189134.97436.e2
72. Young S, Chadwick O, Heptinstall E et al (2005) The adolescent outcome of hyperactive girls. Eur Child Adolesc Psychiatry 14:245–253. https://doi.org/10.1007/s00787-005-0461-z
73. Mikami AY, Hinshaw SP, Lee SS, Mullin BC (2008) Relationships between social information processing and aggression among adolescent girls with and without ADHD. J Youth Adolesc 37:761–771. https://doi.org/10.1007/s10964-007-9237-8
74. Mehta N, Clement S, Marcus E et al (2015) Evidence for effective interventions to reduce mental health-related stigma and discrimination in the medium and long term: systematic review. Br J Psychiatry 207:377–384. https://doi.org/10.1192/bjp.bp.114.151944
75. Thornicroft G, Mehta N, Clement S et al (2016) Evidence for effective interventions to reduce mental-health-related stigma and discrimination. Lancet 387:123–1132